

XRPL Validator — Daily Engineering Report • 2026-08-02

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- 5 commits (xrpl-validator 5)

Engineering changes

- **fix(ledger): remove self-crossed offers off the book tip** (b733ae7 , xrpl-validator) rippled DELETES the taker's own offers off the tip rather than crossing them — BookStep.cpp:415-455 `limitSelfCrossQuality`, which explains the choice itself: "We could skip over the self offer in the book and only cross offers that are not our own. This would make a lot of sense, but we don't do it ... we can only operate on the tip of the order book. We can't leave an offer behind -- it would sit on the tip and ..."
- **fix(ledger): measure a payment hop's carry, don't subtract a sentinel** (7efe2fd , xrpl-validator) An intermediate hop of a multi-hop payment has no real ceiling, so it was handed a sentinel want-cap of (9.99e15, 60) and its output recovered as `want_cap - remaining`. That cannot work. `me_rescale` saturates (`saturating_mul`), so rescaling the sentinel down to a fill's exponent pins it at `u128::MAX` — and it does so again on the NEXT fill, resetting the running remainder and ERASING every fill before it. The ...
- **feat(ledger): AMMDeposit tfTwoAsset equalDepositLimit + tecAMM_FAILED** (a533219 , xrpl-validator) Amount and Amount2 on a two-asset deposit are MAXIMA, not the amounts deposited. rippled derives one side from the other at the pool's ratio and fails the transaction outright when NEITHER derivation fits inside what was offered (AMMDeposit.cpp:721-787): `frac = amount / amountBalance` (Number, 16 digits) `tokensAdj = adjustLPTokens(lpt, multiply(lpt, frac, DOWNWARD))` `frac = tokensAdj / lpt` (`adjustFracByTokens`) `amt2Dep ...`
- **refactor(ledger): one strand code path in Payment** (c858f1f , xrpl-validator) The payment had two execution shapes — a hop chain when Paths named usable intermediate currencies, and a separate direct `cross_engine_to` otherwise. They were the same walk: with no hops the chain is [`spend`, `want`], the loop runs exactly one hop with `last=true`, `benef=dest` and `want_cap=want_gross`, and issues the identical `cross_engine_to` to call the direct branch made. The inflight snapshot iterates `hops` and so is ...
- **feat(ledger): flow the default path alongside Paths, per pass** (99f8a7a , xrpl-validator) rippled ALWAYS builds the default path beside the ones named in Paths (unless `tfNoRippleDirect`), flows each strand, and applies the best-quality PASS each round before re-evaluating (StrandFlow.h:682-805). We ran the specified chain alone.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Oracle-Gated Stop/Trigger Offers** (dex) Add a tfStopOrder flag to OfferCreate plus SFields sfTriggerOracleID (a PriceOracle keylet), sfTriggerPrice, and sfTriggerDirection; a flagged offer i
- **Auditor-Keyed Ciphertext for Permissioned MPT Flows** (audit) Add a new owner-reserved ledger object AuditorKey (ltAUDITOR_KEY, keylet::auditorKey(account)) set by a new AuditorKeySet transaction carrying sfAudit
- **Sponsor-Backed Trust Line Reserves** (fees) Add a new owner-reserved ledger object SponsorAuthorization (ltSPONSOR_AUTH, keylet::sponsorAuth(sponsor, domain)) created by a new SponsorAuthorize t

Appendix — commits

Time	Repo	Commit	Subject
00:49	xrpl-validator	b733ae7	fix(ledger): remove self-crossed offers off the book tip
11:41	xrpl-validator	7efe2fd	fix(ledger): measure a payment hop's carry, don't subtract a sentinel
21:56	xrpl-validator	a533219	feat(ledger): AMMDeposit tfTwoAsset equalDepositLimit + tecAMM_FAILED
22:41	xrpl-validator	c858f1f	refactor(ledger): one strand code path in Payment
23:30	xrpl-validator	99f8a7a	feat(ledger): flow the default path alongside Paths, per pass